

SWEEP rule on the KD-Toric code

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1 SWEEP Rule Decoder on the Toric, version II.

Consider the presentation of a d -dimensional toric by the Cayley graph of the $(\mathbb{Z}_n^d, +)$ group generated by the elements $S = \{1_i : i \in [d]\}$ (where 1_i is the vector consisting of one at the i th coordinate and zero elsewhere). We say that an edge $\{x, y\}$ is positive relative to a plaquette $\{v, g_1v, g_2g_1v, g_2v\}$, defined by a rooted vertex $v \in \mathbb{Z}_n^d$ and the generators $g_1, g_2 \in S$, if $x = v$ and y is either g_1v or g_2v . In our code, bits are set on the plaquette, and the edges correspond to checks that restrict the parity of the bits on their support to be even.

We generalize Toom's rule in the following manner: Each plaquette asks if both of its positive edges point to a syndrome, namely unsatisfied. If so, then the bit flips itself; for convenience, we will think of the vertex at the end of the positive edges as the vertex which makes the decision and flips the plaquette.

Let $\sigma^{(t)}$ be a domain wall at time t , and let $\Gamma^t = \{v : \exists \{v, u\} \in \sigma^t\}$. We denote by ϕ^t the subsets of bits which are flipped as a result of applying the Toom's rule at time t , and by ϕ_v the bits which are flipped by vertex v . We say that u is added to Γ^{t+1} by v at time t , if $u \in \Gamma^{t+1}$ and $u \in V(\phi_v)$. When it clear from the context, we will omit the time, and say that u is added by v .

Claim 1.1. Let $v \in \Gamma^t$ for any u that is added by v , and for any $x \in [d+1]$, we have $L_x(u) \leq L_x(v) + 1$.

Proof. By definition, any u that is added by v is of the form $v + \sum \alpha_j 1_j$, where $\alpha_j \in \{0, 1\}$ and $\sum \alpha_j \leq 2$. Thus, for any $x \in [d]$:

$$L_x(u) = \langle 1_x, v + \sum \alpha_j 1_j \rangle = \langle 1_x, v \rangle + \alpha_x \leq L_x(v) + 1$$

For $x = d+1$ we have:

$$L_x(u) = \langle -\mathbf{1}, v + \sum \alpha_j 1_j \rangle = \langle -\mathbf{1}, v \rangle - \sum_j \alpha_j \leq L_x(v)$$

□

Claim 1.2. Let $v \in \Gamma^t$ such that $L_{d+1}(v)$ is a (local) maxima in Γ^t then for any u which is added by v we have $L_{d+1}(u) \leq L_{d+1}(v) - 1$.

Proof. We will prove that $v \notin \Gamma^{t+1}$, and then by repeating on claim 1.1 where $\sum \alpha_j \geq 1$, we get the desired result. Assume, toward contradiction, that $v \in \Gamma^{t+1}$, and denote by w the vertex that adds v . If $w \neq v$, then there is at least one coordinate i for which $w_i < v_i$, thus $L_{d+1}(w) > L_{d+1}(v)$, in contradiction to $L_{d+1}(v)$ being a maximum in Γ^t . In the other case, where v adds itself, it follows that σ_v can't be decomposed into pairs of unsatisfied checks; namely, $|\sigma_v|$ is odd. We show by induction that impossible, namely that $|\sigma_v|$ is always even. The induction is over the size of the plaquettes subset that induce the boundary σ .

1. Base. Consider a boundary that induced by a single plaquette, in that case it obvious that $|\sigma_v|$ is either zero (where v is not belong to the plaquette) or 2.
2. Assume for any boundary that induced by at most k plaquettes, and consider a boundary σ which is induced by $k + 1$ plaquettes, name this subset F and observes that F can be decomposed to a sum of two disjoint subsets A and B , each with less than $k + 1$ plaquettes, Now:

$$\begin{aligned}\sigma_v &= (\partial A)_v \cup (\partial B)_v \\ \Rightarrow |\sigma_v| &= |(\partial A)_v| + |(\partial B)_v| - 2|(\partial A)_v \cap (\partial B)_v| = \text{even}\end{aligned}$$

Where we used the induction assumption, which tells that $|(\partial A)_v|$ and $|(\partial B)_v|$ are even numbers.

□

Claim 1.3. Let $x \in [d]$ and let $v \in \Gamma^t$ such that $L_x(v)$ is a (local) maxima in Γ^t then for any u which is added by v we have $L_x(u) \leq L_x(v)$.

Proof. Since $L_x(v)$ is maximal, v doesn't see edges that goes from it at the positive x -direction, otherwise denote it by e and denote by w the second vertex at the end of e , w has an x -coordinate which is greater than v 's x -coordinate by 1 and in addition $w \in \Gamma^t$ (since it lays on the support of e). Therefore $L_x(w) > L_x(v)$ in contradiction to $L_x(v)$ be maximal in Γ^t . Hence, any plaquette that is being flipped by v lies on edges associated with 1_j such that $j \neq x$. Thus, any vertex u that is added by v has the form $v + \sum_{j \in [d] \setminus \{x\}} \alpha_j 1_j$, where $\alpha_j \in \{0, 1\}$ and $\sum \alpha_j \leq 2$. So:

$$L_x(u) = \langle 1_x, v + \sum_{j \in [d] \setminus \{x\}} \alpha_j 1_j \rangle = \langle 1_x, v \rangle = L_x(v)$$

□

Claim 1.4. Let Γ^t be the vertices in the support of the domain wall σ^t . Denote $v_i = \arg \max_{v \in \Gamma^t} L_i(v)$. Then for any $i \in [d]$, one can find $u \in \Gamma^{t-1}$ such that u adds v_i and $L_i(u) \geq L_i(v_i)$. In addition, one can find $u_{d+1} \in \Gamma^{t-1}$ such that u_{d+1} adds v_{d+1} and $L_{d+1}(u_{d+1}) = L_{d+1}(v_{d+1}) + 1$.

Proof. Just take u_i to be any vertex that adds v_i . By claims claim 1.1, claim 1.2, and claim 1.3, we get the claim. □

[COMMENT] $V(\phi_v)$ was not defined. $v \leq w$ also not defined.